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G1603, SAMARTH GARDEN, BHANDUP (W), MUMBAI 400078, MAHARASHTRA.
9137815600 / 9029767757
pm.team@arcaisys.com

DRISHTI Innovation Challenge

PARINAM

Platform for Analysis of Related Impacts in Naval Assembly Modifications

Powered by CATALYST Architecture

Proposed Technical Solution · Annexure 2

PROBLEM STATEMENT-70

Field	Details
Applicant Name	Leejo Purvathan
Company Name	Arcaisys
Challenge	PROBLEM STATEMENT-70
Duration	20 Months
Contact Email	pm.team@arcaisys.com



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9137815600 / 9029767757
pm.team@arcaisys.com

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5. Proposed Technical Solution

PARINAM is structured as a unified, six-layer, fully deterministic and AI-augmented architecture. The system is designed to be primarily deterministic architecture with Intelligence components. Every layer has well-defined interfaces enabling independent module validation and incremental integration. The system operates entirely on-premise and requires no external network connectivity, satisfying defence-grade air-gap requirements. The system is designed to operate reliably under limited data conditions, ensuring that all core functionalities remain fully operational without dependence on machine learning models.

PARINAM is built on the **CATALYST architecture framework (Change Analysis Through Linked Systems and Intelligent Tracking)**, which serves as the underlying engine layer enabling dependency modelling, impact propagation, constraint validation, and estimation functions.

The CATALYST framework is a modular, engine-based architecture that encapsulates all core computational capabilities of the system, including graph-based reasoning, rule evaluation, and estimation engines. It is not a standalone product, but a reusable technical foundation upon which PARINAM is implemented.

5.1 Digital Twin Data Model

Each ship component is represented as a structured data node within a unified component model.

The node schema is:

- component_id: UUID , globally unique identifier, linked to both CAD assembly path and BOM part number
- geometry_features: {bounding_box, surface_area, volume, centroid, moments_of_inertia, topology_hash}
(**OPTIONAL**: Applicable if CAD Data is available)
- material: {material_code, grade, density, yield_strength, UTS, thermal_conductivity, corrosion_class}
- mass_properties: {mass_kg, cog_xyz, inertia_tensor} (**OPTIONAL**: Applicable if CAD data is available or from BOM listed mass details)
- bom_entry: {part_number, description, quantity, unit_cost, supplier_code, lead_time_days}
- rule_bindings: [{rule_id, class_society, parameter_name, limit_value, limit_type}]
- version_history: [{timestamp, changed_by, change_description, delta_hash}]

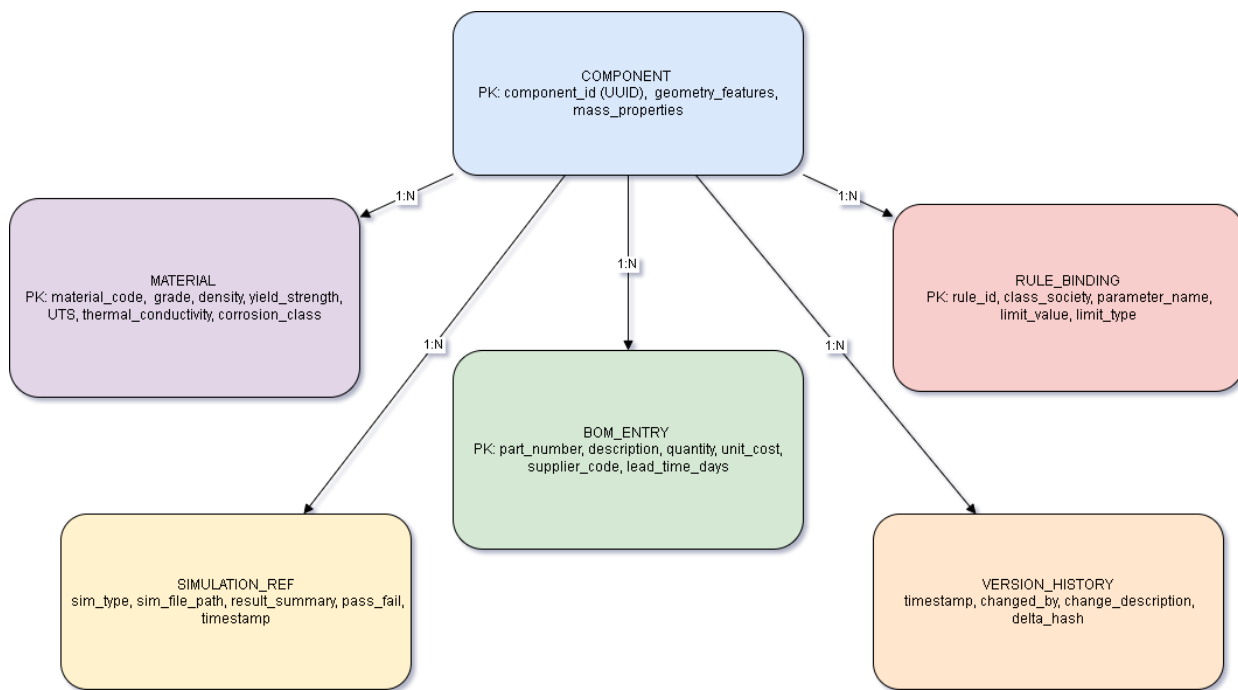


Figure: Digital Twin Core Data Model , Entity Relationship Diagram

5.2 NLP Input Processing Engine

The system accepts two input modalities: (a) natural language design change requests via the dashboard or API; and (b) structured CAD model difference files parsed by the CAD diff adapter. The BERT NLP model is fine-tuned on a domain-specific corpus of historical naval design change request documents.

Named Entity Recognition (NER) Schema

Entity Tag	Definition	Example
COMPONENT	Primary component affected	Cable Tray CT-41A, Bulkhead BH-107
CHANGE_TYPE	Nature of change	relocation, resize, material substitution, removal, addition, rerouting
MAGNITUDE	Quantified extent of change	3 metres, +15% cross-section, Frame 32 to Frame 35
LOCATION	Physical zone reference	Deck 2, Compartment E-14, Frame 32–35
AFFECTED_ATTRIBUTES	Properties being altered	routing_path, cable_count, tray_capacity, load_rating



- Output: A validated JSON change object conforming to PARINAM Change Schema v1.0, containing: change_id, component_id, change_type, magnitude, location, affected_attributes[], and confidence_score.

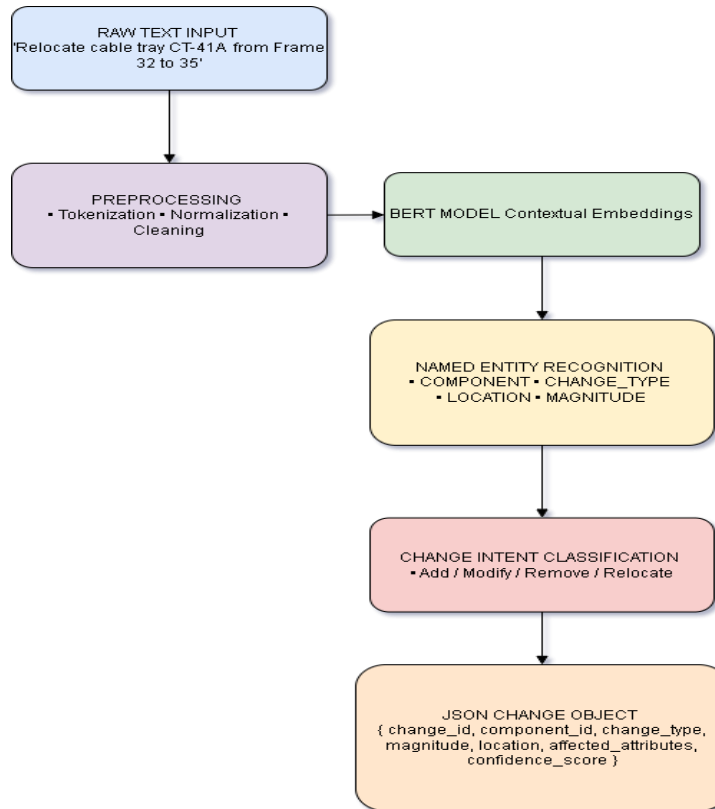


Figure: BERT NLP Processing Pipeline , From Raw Text to JSON Change Object

5.3 Dependency Knowledge Graph Engine

The entire ship engineering system is represented as a Directed Weighted Graph $G = (V, E, W)$ stored in Neo4j. This constitutes the single source of truth for all inter-system dependency reasoning.

Initial graph construction will be validated and refined through Domain Expert inputs to ensure engineering accuracy and reliability of dependency relationships.

Edge Weight Calibration: Initial edge weights are set during Phase 0 through two sources:

(a) SME workshops where domain engineers assign weights based on known coupling strengths (e.g., a cable tray directly inside a structural bay receives STRUCTURALLY_SUPPORTS weight = 0.9)

(b) historical co-change frequency from MDL change records, pairs of components that changed together in >50% of historical changes receive higher weights. Weights are stored as properties on each Neo4j relationship and are recalibrated quarterly as new change records accumulate.



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9137815600 / 9029767757
pm.team@arcaisys.com

Graph Construction Pipeline

- BOM XML/CSV ingested into component registry , each line item becomes a node with attributes: part_number, system, material, location, unit_cost, lead_time
- ASSEMBLY_CHILD edges populated from BOM hierarchy (CSV/XML parent-child fields)
- PHYSICAL_INTERFACE edges populated from SME dependency matrix (with OCC as optional Phase 6 enhancement)
- FUNCTIONAL_DEPENDENCY and RULE_GOVERNS edges populated from the rule catalog and SME-maintained dependency matrix

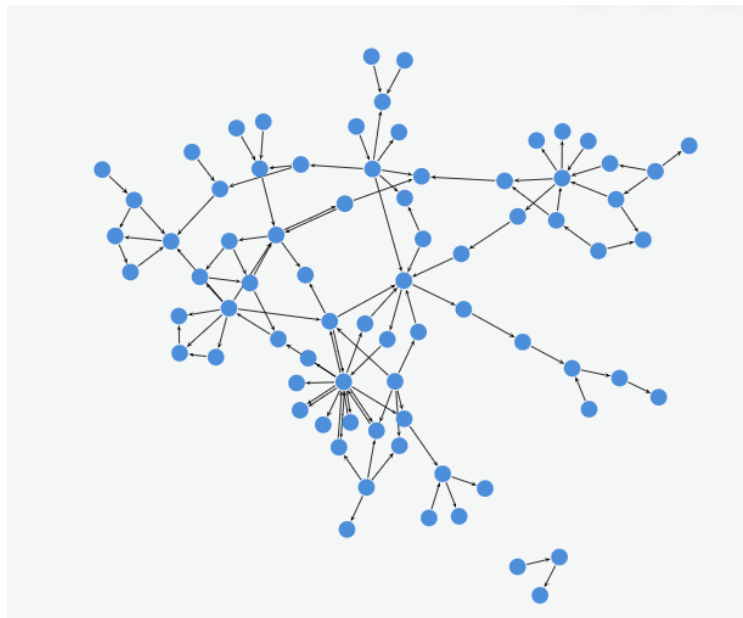


Figure: Neo4j Dependency Knowledge Graph

Edge Type Taxonomy

Edge Type	Physical Meaning	Weight Basis
STRUCTURALLY_SUPPORTS	Component A bears load of Component B	Load-sharing ratio
ELECTRICALLY_FEEDS	Power or signal dependency between components	Voltage/current dependency ratio
THERMALLY_ADJACENT	Thermal influence between co-located components	Spatial proximity factor
ROUTED_THROUGH	Cable or pipe routed through a structural element	Contact area ratio
MECHANICALLY_CONNECTED	Direct mechanical fastening or joint	Contact area / joint stiffness
RULE_GOVERNS	Classification rule applicable to a component	Rule criticality weight



5.4 Impact Propagation Engine

The propagation engine executes a modified Breadth-First Search on the dependency graph starting from the changed component.

The propagation results are fully traceable, with each impacted component linked to its originating dependency path for engineer validation.

Propagation Formula

$$\text{Impact}(n) = \text{Impact}(\text{parent}) \times \text{EdgeWeight}(\text{parent} \rightarrow n) \times e^{(-\lambda \times \text{depth})}$$

- $\text{Impact}(\text{root}) = 1.0$, the changed node always has full impact
- $\text{EdgeWeight} \in [0, 1]$, calibrated per-edge; high weight = strong coupling
- Depth decay: $e^{(-\lambda \times \text{depth})}$, $\lambda = 0.2$, prevents low-relevance nodes in reports
- Threshold $\theta = 0.05$, nodes with $\text{Impact}(n) < \theta$ are pruned from output
- Multi-path accumulation: if reachable via multiple paths, score = maximum across all paths (conservative upper bound)
- Configurable depth limit: default 6 hops; tunable per project type

Sensitivity Classification Thresholds:

- HIGH: Impact score ≥ 0.40 — mandatory engineer review before change approval
- MEDIUM: Impact score $0.15\text{--}0.39$ — flagged for awareness; review recommended
- LOW: Impact score $0.05\text{--}0.14$ — logged; no mandatory review action required
- PRUNED: Impact score < 0.05 (θ threshold) — excluded from output report

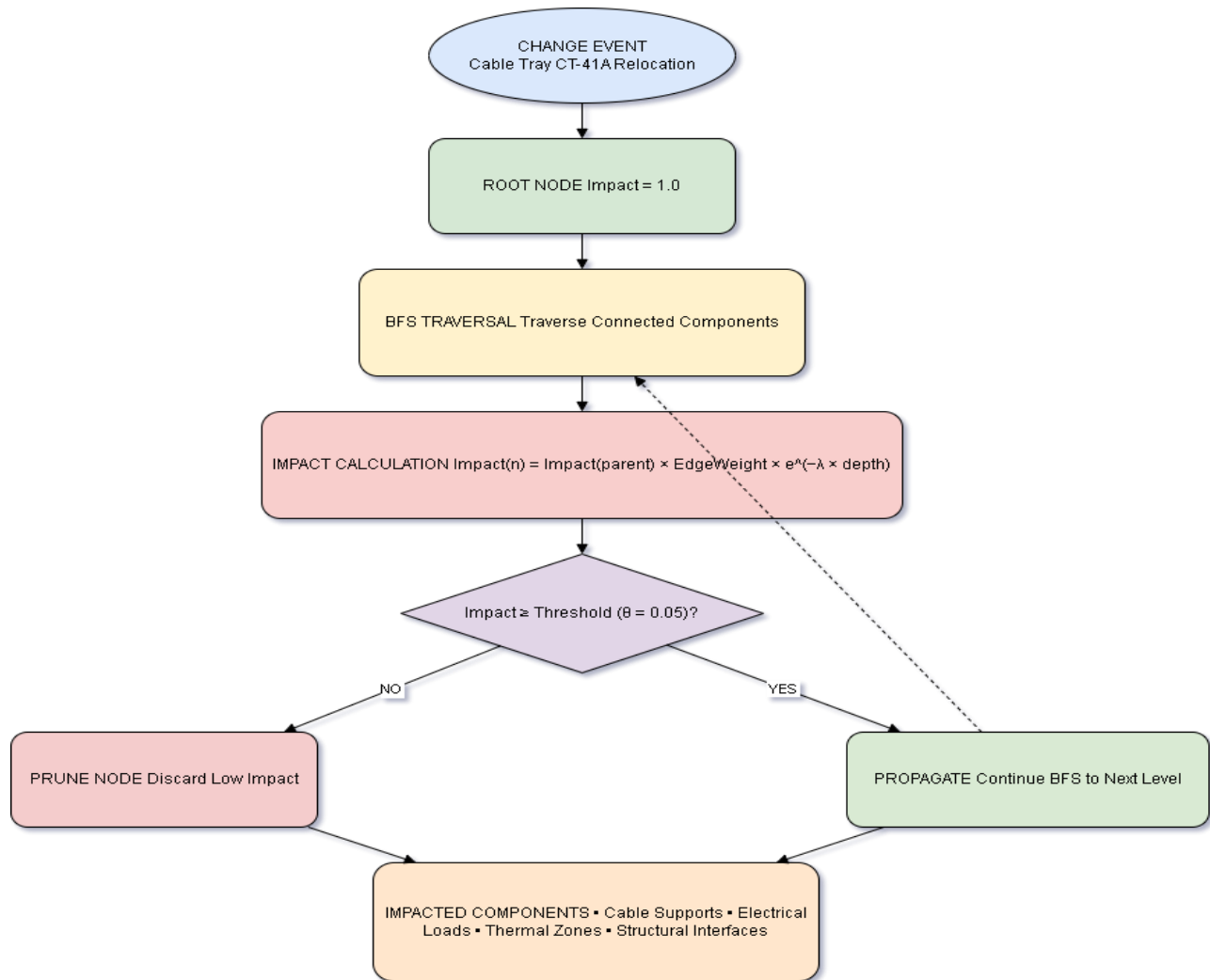


Figure: Impact Propagation BFS Flow , Example: Cable Tray Relocation Cascade



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9137815600 / 9029767757
pm.team@arcaisys.com

5.5 PARINAM System (Input + CATALYST Engine + Output)

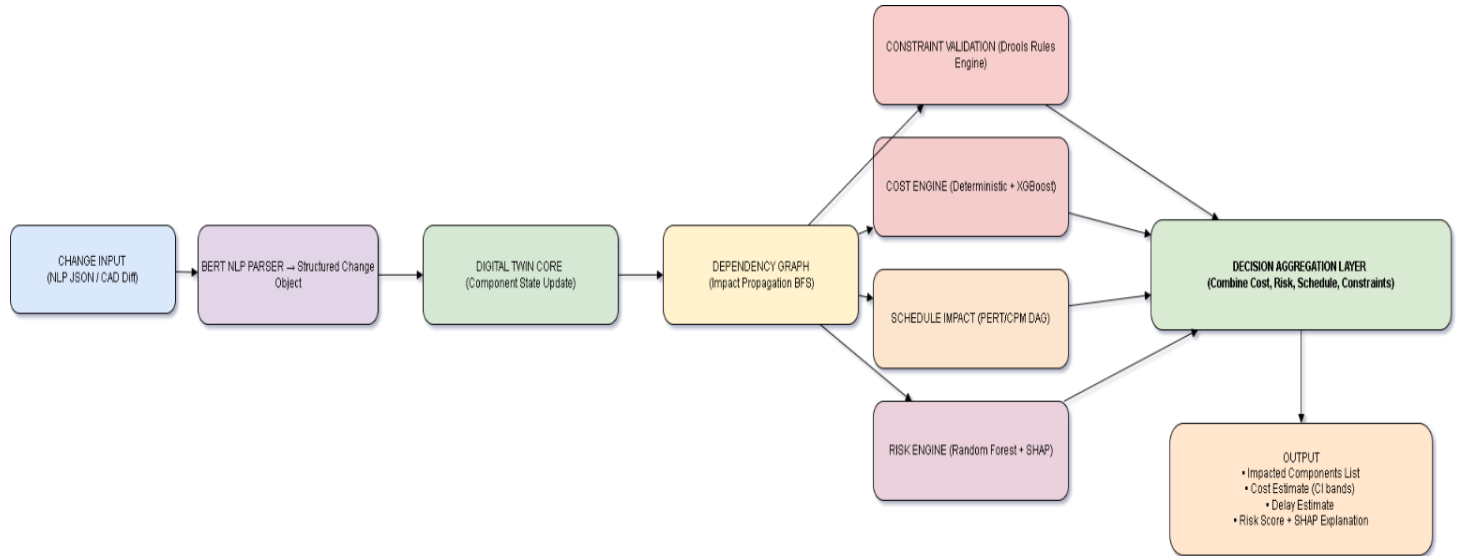


Figure: CATALYST Engine Stack , End-to-End Data Flow Diagram

5.5.1 Constraint Validation Engine

Constraint Category	Rule Example	Validation Logic	Severity
Cable Tray Capacity	Fill ratio $\leq 40\%$	$(\text{current CSA} + \text{new cables CSA}) / \text{tray_capacity} \leq 0.40$ per IEC 61084-1	CRITICAL
Structural Load	Deflection $\leq L/250$	$\text{distributed_load} / \text{section_modulus} \leq \text{allowable_stress} \times \text{SF}$ (SF=1.5)	CRITICAL
Thermal Clearance	Min 100mm from heat source	Euclidean distance (cable centroid, heat source envelope) $\geq 100\text{mm}$	HIGH
Bend Radius	Min $\geq 12 \times \text{OD}$	$\text{route_bend_radius} \geq 12 \times \text{cable_outer_diameter}$ per IEC 60092	HIGH
Fire Zone Segregation	No co-routing across fire zones	Cable does not cross fire zone boundary without fire-stop fitting	CRITICAL
Pressure Rating	Pipe rating $\geq 1.5 \times \text{MAWP}$	$\text{selected_pipe_rating} \geq 1.5 \times \text{MAWP}$ per system spec	CRITICAL
Access Clearance	600mm maintenance aisle	New component does not reduce nearest maintenance aisle below 600mm	MEDIUM



5.5.2 Two-Layer Cost Estimation Engine

- The cost estimation engine is structured as a deterministic primary layer supported by an intelligent refinement layer.
- The deterministic layer computes cost based on BOM, labour, rework, and indirect factors and remains fully functional independent of historical datasets.
- Layer 1 , Deterministic Formula: Total Cost = Material Cost + Labour Cost + Rework Cost + Indirect Costs ± Variance for computing the Cost range. Each component priced using rate tables and BOM unit costs.
- Variance is calculated using a rule-based uncertainty factor table (change_type × system_domain → uncertainty range 10–30%)
- The system maintains a variance tracking log that computes historical estimate-vs-actual delta per change type and system domain. This is surfaced on the dashboard as a 'variance forecast indicator' alongside each cost estimate.
- Layer 2 , XGBoost ML Model: Trained on historical naval design change records. Features include impact_score, affected_system_count, constraint_violation_count, change_type, and magnitude. Output: predicted total cost with 80% and 95% confidence intervals.
- Both outputs are presented side-by-side for triangulation , engineers see both the deterministic estimate and the ML prediction, with CI bands.

5.5.3 PERT/CPM Schedule Impact Module

- Operates on a project schedule Directed Acyclic Graph (DAG) constructed from the programme Gantt.
- For each impacted task: $TE = (O + 4M + P) / 6$, optimistic, most likely, and pessimistic durations from SME estimates.
- Critical path computed using NetworkX; affected critical path nodes highlighted with float consumption per task.
- Output: estimated delay in working days/weeks, recovery action recommendations.
- Recovery actions are drawn from a pre-built library and presented with feasibility flags.

Examples:

- (a) Overtime authorisation: applicable if delay ≤ 5 days and resource availability permits
 - (b) Parallel working: tasks with no dependency conflict rescheduled to run concurrently
 - (c) Scope reduction: optional work packages de-prioritised to recover float
 - (d) Subcontracting: flagged when in-house capacity is the bottleneck.
- Each option includes estimated days recovered and resource cost impact.



5.5.4 Risk Scoring Engine

- Risk scoring is primarily derived from deterministic indicators such as impact propagation severity, constraint violations, and affected system breadth.
- Where sufficient historical data exists, Intelligent models will be used to provide additional insights, but are not required for system operation.
- Random Forest classifier trained on historical overrun data. Features: impact_severity, overrun_probability, constraint_violation_count, propagation_breadth.
- Output: composite risk score 0.0–1.0 with Low/Medium/High classification.
- SHAP explainability layer outputs top 3 risk drivers in plain English for each assessment.
- Mode 1 (Day 1, no ML data): deterministic risk score based on violation count, impact breadth, and system criticality
- Mode 2 (Post-pilot, sufficient data): Random Forest augments the deterministic score

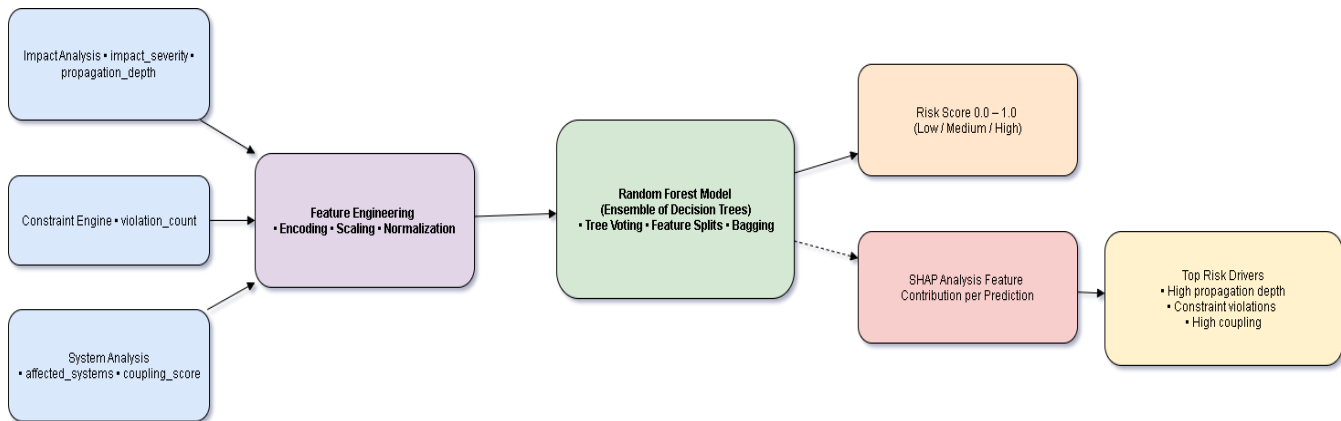


Figure: CATALYST Risk Score Computation , Random Forest Feature Importance Diagram

5.5.5 Unified System Dashboard

The Decision Intelligence Dashboard is a secured, role-based web application deployed entirely on-premise (React + TypeScript). It serves as the single operational interface for engineers, analysts, and programme managers to submit change requests, monitor assessment progress, and act on outputs.

Dashboard Panels and Components:

1. Change Request Submission Panel Accepts free-text or structured design change input. NLP parser processes natural language input and returns the extracted JSON change object for engineer confirmation before analysis is triggered. Structured input form available as fallback.

2. Interactive Dependency Graph Canvas D3.js or Cytoscape.js rendered graph showing the changed component as root node with all propagation paths. Nodes are colour-coded by sensitivity class (High = red, Medium = amber, Low = green). Edge weights displayed on hover. Engineer can click any node to drill into its component profile, impact score, and propagation path from root.



3. Impacted Systems Summary Panel Tabular and visual breakdown of impacted components grouped by system domain (Structural, Electrical, Piping, HVAC, Propulsion). Shows component count, aggregate impact score, and highest violation severity per system. Allows filtering by system or sensitivity class.

4. Constraint Violations Panel Lists all constraint violations detected across impacted components. Each finding shows: Rule ID, standard reference (e.g., IEC 61084-1), component name, actual value, limit value, severity (Critical / High / Medium), and recommended action. Sortable by severity. Critical violations highlighted in red with a mandatory acknowledgement flag before approval can proceed.

5. Cost Estimate Card Side-by-side display of the deterministic cost estimate and the ML (XGBoost) cost prediction. Deterministic estimate shows line-item breakdown: Material, Labour, Rework, Indirect, and Dry-Dock Premium. ML estimate shows predicted total with 80% and 95% confidence interval bands. A variance forecast indicator flags if this change type has historically exceeded estimates by a material margin.

6. Schedule Impact View Gantt-style view of affected tasks with delay highlighted on the critical path. Shows estimated delay in working days, float consumption per affected task, and tasks flagged as near-critical (float < 3 days). Recovery action recommendations listed below the Gantt.

7. Risk Score Gauge Circular gauge displaying the composite risk score (0.0–1.0) with Low / Medium / High band demarcation. Below the gauge: top 3 SHAP-derived risk drivers displayed in plain English (e.g., "Risk is HIGH primarily due to: [1] Critical cable tray violation at CT-41A, [2] 14 nodes impacted in Electrical Distribution, [3] Schedule float exhausted on Task E-041").

8. Approval Workflow Panel Displays the current approval status of the change request. Approver role can accept, reject, or request revision of the assessment with a recorded reason. All approval actions are written to the immutable audit log with timestamp and user ID.

9. PDF Report Export One-click generation of a full assessment report in PDF format containing all panels above in a structured layout, with cover page, change request summary, impacted systems map, violation table, cost estimate, schedule impact, and risk score. Report is auto-named with change_id and timestamp.

10. Audit Trail Viewer (*Admin / Auditor role only*) Read-only log of all change requests, engine outputs, user actions, and approval decisions. Filterable by date range, user, change type, and risk category. Each record includes a hash chain reference for tamper-evidence verification.

5.6 Integration Layer & API Architecture

All engine interactions are orchestrated by a FastAPI + Uvicorn async backend exposing a REST API gateway.

Integration Target	Protocol	Notes
PLM Systems (Teamcenter, Windchill)	REST Adapter Stubs	Adapter stubs provided; full integration activated on client environment access
ERP Systems (SAP PM, Oracle)	REST / OData	BOM sync and cost rate table ingestion; ERP-side configuration required
CAD Systems (CATIA, NX, SolidWorks)	File Watcher / Webhook	On-save hooks trigger CAD ingestion pipeline; webhook endpoint configurable per format



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pm.team@arcaisys.com

Integration Target	Protocol	Notes
Project Management (Primavera P6)	XML Export	Schedule DAG exported in P6 XER/XML format; import adapter for baseline schedule ingestion
Async Task Queue (Celery + Redis)	Internal	All long-running jobs (ML inference) dispatched via Celery; status polling via REST

Due to restricted access to MDL internal systems during the proposal phase, integration components are designed as schema-agnostic adapters, with full system integration to be executed post-award based on available interfaces.

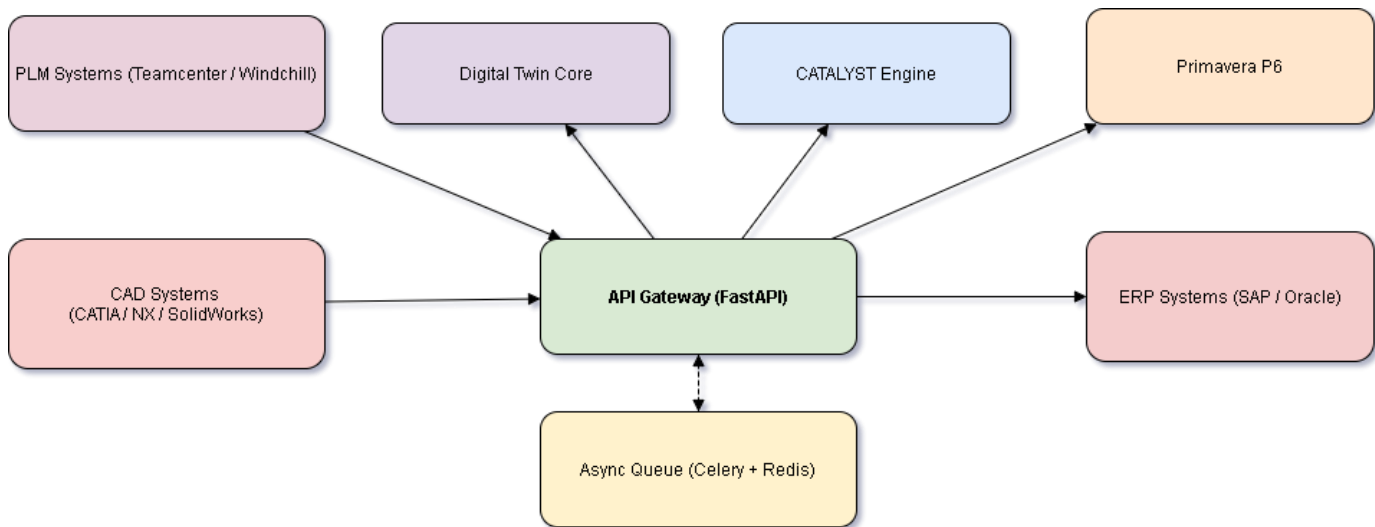


Figure: System Integration Architecture , API Gateway and Connector Topology

5.7 Security & Deployment Architecture

- Deployment: Fully containerized via Docker-Compose. Air-gap compatible, no external network required post-deployment.
- Role Based Access Control: Four roles , Viewer, Analyst, Approver, Admin. Role assignments managed via on-premise LDAP/AD integration.
- Audit Trail: Full immutable audit log of all change requests, engine outputs, user actions, and approval decisions stored in PostgreSQL with hash-chaining.
- Phase 6 Enhancement: Zero-trust architecture (mTLS between microservices); RBAC combined model.
- Data Encryption: All data at rest encrypted using AES-256 (Neo4j database, PostgreSQL tablespace, MinIO bucket). All inter-service communication encrypted using TLS 1.3. ML model artifacts and audit logs stored in encrypted volumes. No plaintext sensitive data persisted at any layer



5.8 Challenges & Mitigation

Challenge	Risk	Mitigation
Data Scarcity for ML Training	Insufficient historical change records to train cost/risk models with high accuracy in early phases	Two-layer architecture: deterministic formula active from Day 1; ML layer activated when sufficient validated data is available; transfer learning from publicly available naval datasets
Graph Construction Quality	Incomplete or inaccurate BOM-to-dependency mappings produce false negatives in impact propagation	Phase 0 constraint catalogue; SME-reviewed seed templates for standard vessel architectures; $\geq 95\%$ coverage gate before go-live
Classification Rule Freshness	Naval classification rules updated periodically; stale rules cause false compliance passes	Structured rule catalog with effective_date fields; quarterly rule update process defined in FOM; IRS/DNV/BV rule feeds monitored
Air-Gap Constraint	On-premise deployment complicates model retraining and software updates	Containerized deployment with offline update packages; retraining pipeline included in handover; all model artifacts version-controlled in MinIO

6. Capabilities & Competencies

Arcaisys brings a focused combination of applied AI, software engineering, and defence-grade system design expertise directly relevant to PARINAM. Key competencies:

- Full-stack AI/ML development: proficiency in PyTorch, XGBoost, BERT fine-tuning, and SHAP explainability for production-grade ML systems
- Graph database engineering: practical experience with Neo4j for knowledge graph construction, Cypher query optimization, and BFS traversal at scale
- Backend engineering: FastAPI, PostgreSQL, Celery, microservices architecture with async job dispatching and REST API design
- Frontend development: React + TypeScript for engineer-facing dashboards with real-time data visualization, RBAC, and PDF report generation
- Security engineering: implementation of RBAC, audit trail systems; familiarity with zero-trust architecture principles
- Naval domain research: engagement with IRS, DNV, Lloyd's Register classification rule structures; engineering constraint modelling for cable trays, structural members, and piping systems
- Technical communication: authored detailed project reports, system architecture documentation, and API specifications for complex multi-module platforms
- Our technical solutions have undergone rigorous expert validation by senior faculty at CDAC-Mumbai. We are Looking forward to collaborating with them.



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9137815600 / 9029767757
pm.team@arcaisys.com

Request for Document Verification & Problem Statement Alignment – iDEX DISC (PS-70) Submission

From pm.team@arcaisys.com

kiranw@cdac.in

May 1, 6:33 PM

Dear Madam,

I hope you are doing well.

We are writing on behalf of **Arcaisys Pvt. Ltd.**, a Mumbai-based startup, seeking your guidance regarding our participation in the **iDEX DISC Challenge – Problem Statement 70 (AI-Based System for Design Change Impact Analysis and Cost Estimation)**.

We have developed our proposed solution titled **"PARINAM – Change Analysis Through Linked Systems and Intelligent Tracking"**, which is designed as a data-driven and AI-augmented platform to address the challenges outlined in the problem statement. The system focuses on:

- Predicting cross-disciplinary impact of design changes using a dependency knowledge graph
- Generating intelligent cost estimates with confidence ranges
- Estimating schedule impact using PERT/CPM-based models
- Processing change requests using NLP-based document understanding
- Providing a unified decision-support dashboard for management

As highlighted in the problem statement, current processes are manual, fragmented, and lead to cost overruns and delays. Our solution aims to address these gaps through a structured, scalable, and partially AI-driven approach aligned with MDL's operational requirements.

We have attached the following documents for your review:

1. Proposed Solution Document (Annexure 1)
2. Technical Solution Document (Annexure 2)

We kindly request your support in:

1. **Verifying whether our solution is aligned with all aspects of Problem Statement-70**, particularly:

- Design change impact propagation across systems
- Cost estimation methodology and confidence modelling
- NLP-based document understanding capability
- Dashboard and decision-support requirements

2. **Reviewing the attached documents for:**

- Technical completeness and clarity
- Any gaps or improvements required prior to final submission

Your expert feedback will be extremely valuable in helping us refine our proposal and ensure alignment with CDAC and iDEX expectations.

We would be grateful if you could kindly review the documents at your convenience and share your inputs.

Thank you for your time and support.

Warm regards,

Leejo Xavier Puravathan

Arcaisys Pvt. Ltd.

Mumbai, Maharashtra

+91-9029767757

pm.team@arcaisys.com

2 MB 3 attachments Download all

PARINAM...70.pdf 525 KB

PARINAM_Pr...70.pdf 1 MB

1. AI-Base...70.pdf 186 KB



7. Case Study for the Proposed System

1. Scenario Overview

A naval design engineer submits a design change request:

"Relocate Cable Tray CT-41A from Frame 32–35 to Frame 36–40 due to spatial conflict with HVAC ducting."

This type of change typically causes multi-domain cascading effects, including:

- Electrical rerouting
- Structural support adjustments
- Compliance risks (IEC cable tray fill, bend radius)
- Schedule disruption (installation dependencies)
- Cost escalation due to rework

Traditional workflows would require manual coordination across multiple engineering teams, often leading to delays and inconsistent assessments. PARINAM replaces this with a unified dashboard-driven analysis pipeline.

2. Step-by-Step Dashboard Execution Flow

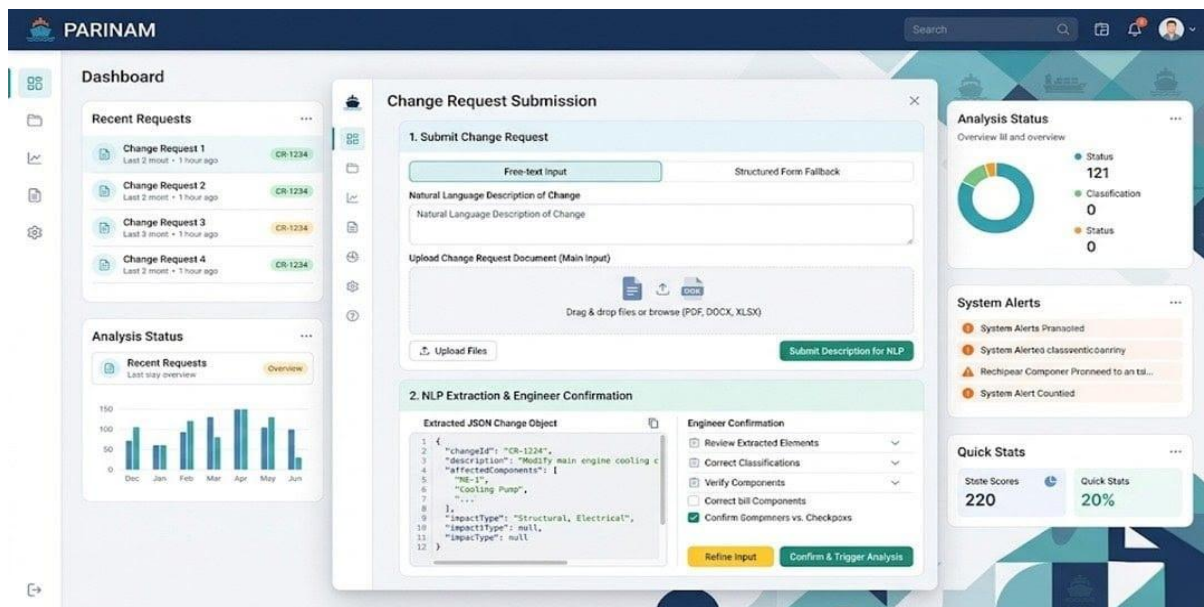
1: Change Request Submission Panel

The engineer inputs the change via natural language.

- The BERT-based NLP engine extracts:
 - Component: CT-41A
 - Change Type: Relocation
 - Magnitude: Frame 32–35 → 36–40
- Output: Structured JSON change object
- Engineer validates before triggering analysis

Outcome:

Eliminates ambiguity and ensures standardised machine-readable input.





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2: Interactive Dependency Graph Canvas

- The system places CT-41A as the root node
- Executes BFS-based propagation on Neo4j graph
- Displays:
 - Connected systems (Electrical, Structural, HVAC)
 - Edge weights (dependency strength)
 - Sensitivity classification:
 - High (Red)
 - Medium (Amber)
 - Low (Green)

Example Insight:

- Electrical distribution nodes show high impact (≥ 0.40)
- Structural supports show medium impact
- HVAC interaction nodes appear due to spatial adjacency

Outcome:

Provides visual traceability of cascading impact paths, which is otherwise impossible in manual workflows.



3: Constraint Violations Panel

Detected violations include:

- Cable tray fill ratio > 40% (IEC 61084-1)
- Bend radius violation in rerouted cables
- Thermal clearance breach near HVAC duct



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Each violation includes:

- Rule ID
- Actual vs permissible values
- Severity classification
- Recommended corrective action

Critical Feature:

- Mandatory acknowledgement required before approval

Outcome:

Ensures compliance enforcement is not bypassed, unlike manual review systems.

PARINAM ENG. OFFICER: L. SMITH

CRITICAL ENGINEERING CONSTRAINT VIOLATION ALERT

MANDATORY ACKNOWLEDGEMENT REQUIRED

Rule ID	Standard (IEC 61084-1)	Component Name	Actual Value	Limit Value	Severity	Recommended Action
R-001	61084-1:4.2	CT-41A - Power Trunk	Frame 32.5	Frame 32.0 - 40.0	Critical	Initiate Reroute
R-002	61084-1:4.2	CT-41A - Power Trunk	Frame 32.5	Frame 32.0 - 40.0	Critical	Initiate Reroute
R-003	61084-1:4.2	CT-41A - Power Funnot	30%	30%	Critical	Initiate Reroute
R-105	61084-1:6.1	PS-A	95% Load	80% Load	High	Load Shedding Plan
R-104	61084-1:6.1	PS-A - Power Funnet	95% Load	80% Load	High	Load Shedding Plan
R-105	61084-1:6.1	PS-A	95% Load	80% Load	High	Load Shedding Plan
R-106	61084-1:6.1	PS-A - Range	20%	20%	High	Load Shedding Plan
R-107	61084-1:6.1	PS-A - Serner	95% Load	80% Load	High	Load Shedding Plan

CONFIRM & RUN ANALYSIS

4: Cost Estimate Card

Two-layer estimation:

Deterministic Output: This is calculated as cumulation of Material, Labour, Rework, Indirect and Dry-Dock premium costs

ML Prediction (XGBoost): Predicts using Regression and 85% and 95% confidence limits

Variance Indicator:

- Similar historical changes show +18% overruns

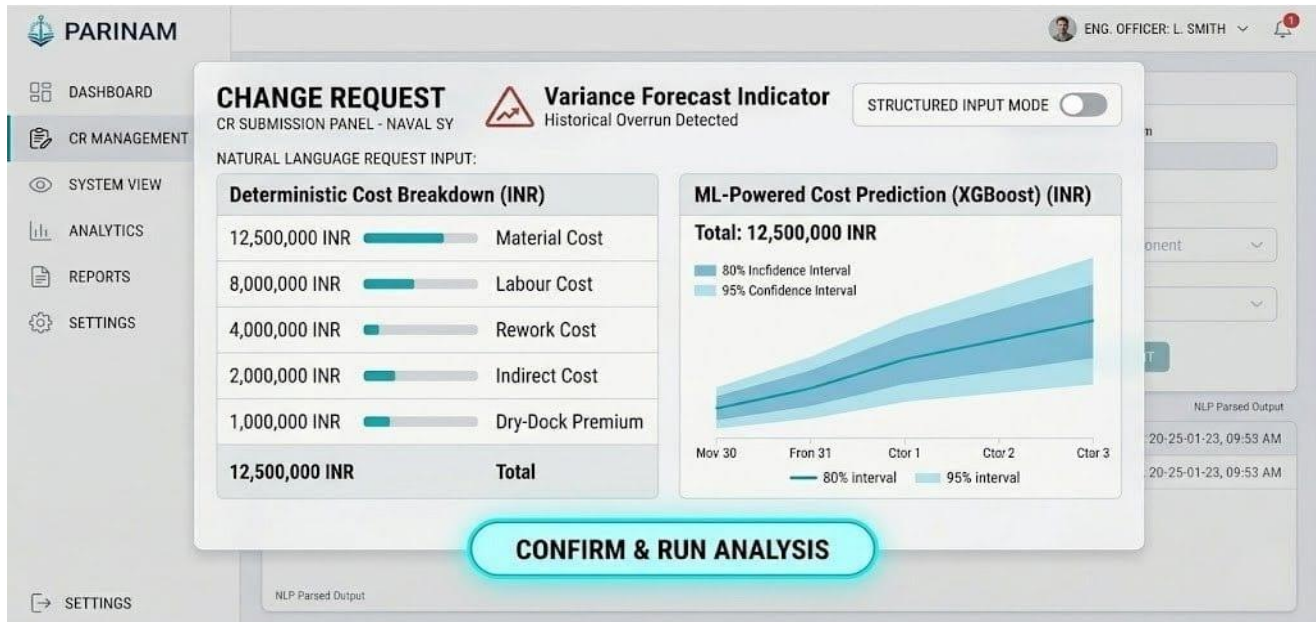
Outcome:

Provides decision confidence through triangulation, not single-point estimates.



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pm.team@arcaisys.com



5: Schedule Impact View

- Affected tasks highlighted on Gantt chart
- Critical path recalculated using PERT/CPM

Findings:

- Delay: +6 working days
- Float exhausted on Electrical Installation Task E-041
- Near-critical tasks identified (<3 days float)

Recommended Recovery Actions:

- Parallel routing execution
- Overtime allocation (feasible)
- Partial subcontracting

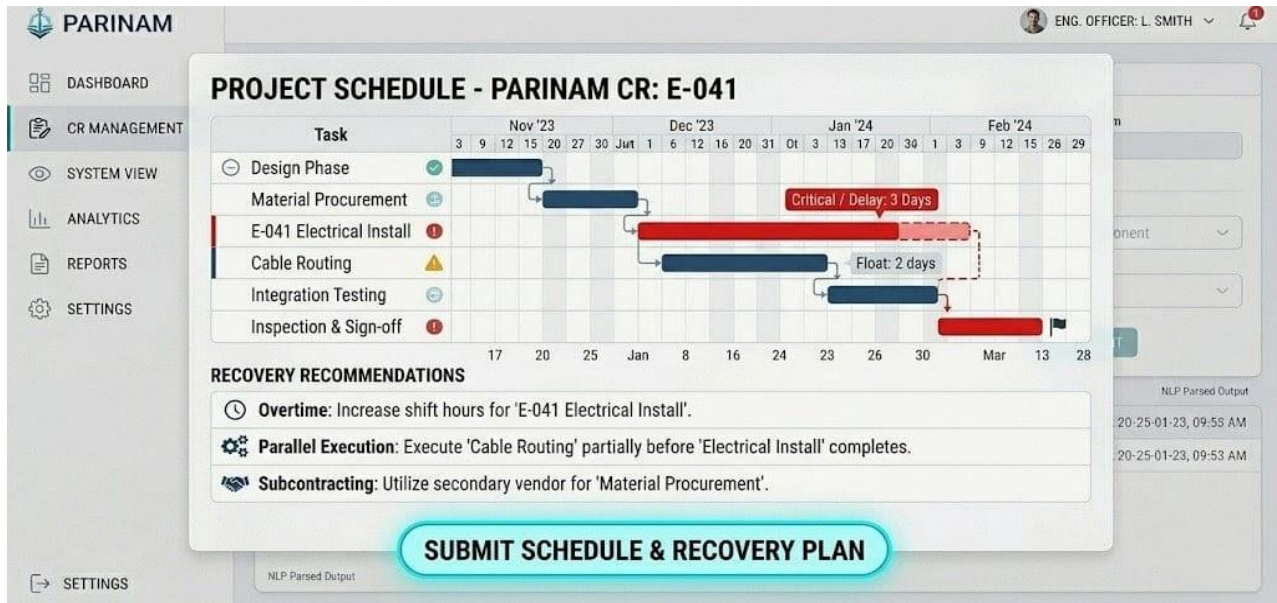
Outcome:

Transforms schedule analysis from static planning to dynamic impact simulation.



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9137815600 / 9029767757
pm.team@arcaisys.com



6: Risk Score Gauge

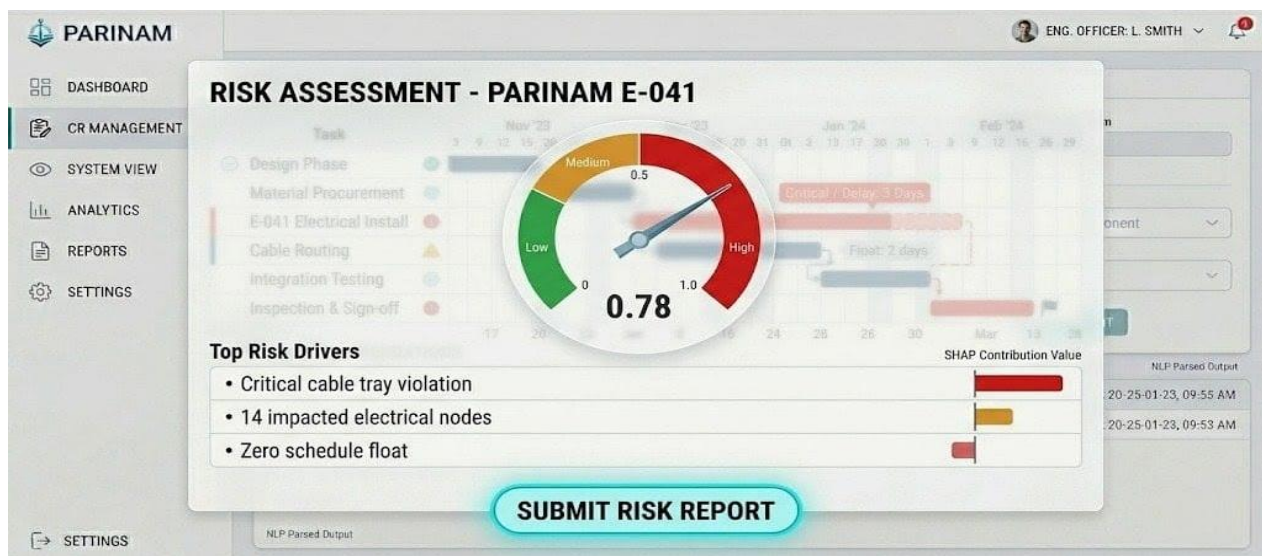
- Composite Risk Score: 0.78 (HIGH)

Top SHAP Drivers:

- Critical cable tray violation
- High propagation breadth (14 nodes in Electrical)
- Critical path delay with zero float

Outcome:

Delivers explainable AI-backed risk justification, not black-box scoring.





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G1603, SAMARTH GARDEN, BHANDUP (W), MUMBAI 400078, MAHARASHTRA.
9137815600 / 9029767757
pm.team@arcaisys.com

7: Audit Trail Viewer

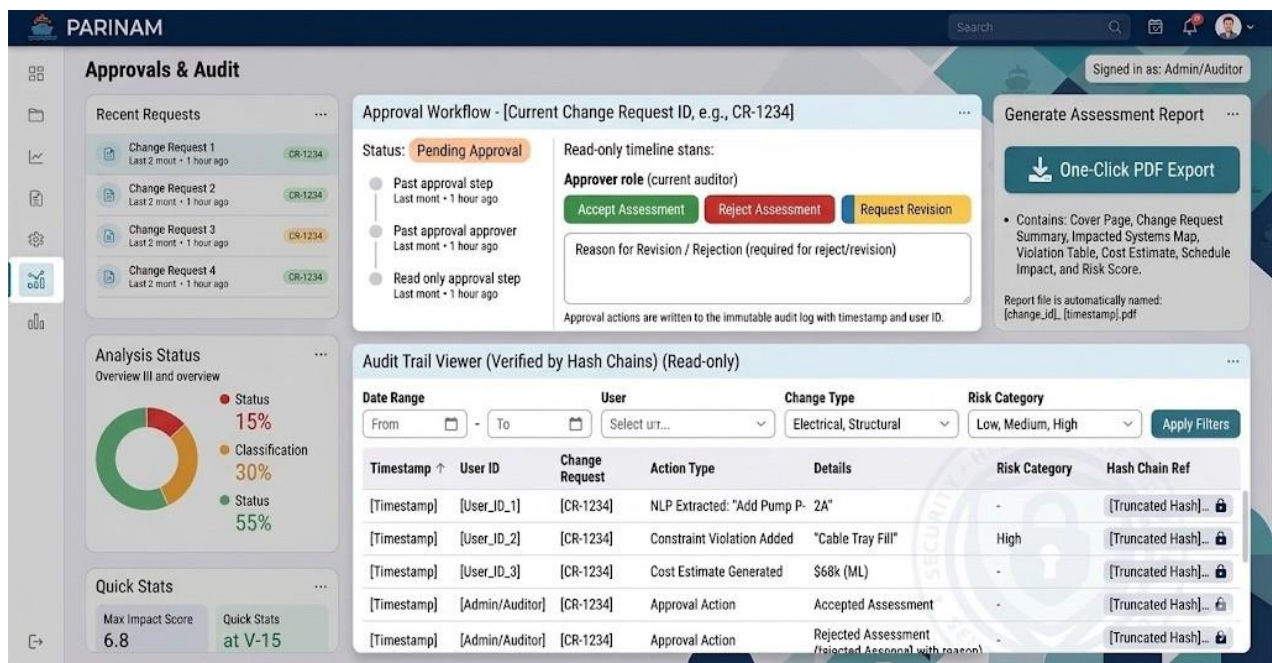
- Full trace of Inputs , Engine outputs, User decisions

Includes:

- Hash-chained records (tamper-evident)
- Filter by: Date, User, Risk category

Outcome:

Enables defence-grade auditability and compliance verification.



Disclaimer

Please note that the dashboard designs and interfaces showcased in this case study are conceptual representations. They are intended solely to illustrate the proposed features, user interface, and workflow of the PARINAM Change Impact Analysis System. These images serve as a visual aid for the case study and do not reflect the finalized, production-ready software product.



8. List of Abbreviations & Technical Terms

The following abbreviations and technical terms are used throughout this document.

Abbreviation / Term	Definition / Explanation
AD	Active Directory , Microsoft's directory service used for identity management and user authentication in enterprise environments.
AI	Artificial Intelligence , Computational systems that perform tasks typically requiring human intelligence, such as pattern recognition, decision-making, and natural language understanding.
API	Application Programming Interface , A defined set of rules and protocols allowing software components to communicate with each other.
BERT	Bidirectional Encoder Representations from Transformers , A pre-trained NLP deep learning model developed by Google; used here for parsing and classifying natural language design change requests.
BFS	Breadth-First Search , A graph traversal algorithm that explores nodes layer by layer from a starting point; used in CATALYST's impact propagation engine to identify all affected components.
BOM	Bill of Materials , A structured list of all components, assemblies, sub-assemblies, raw materials, and quantities required to manufacture or assemble a product (here, a naval vessel).
BV	Bureau Veritas , An international classification society that sets and enforces technical and safety standards for ships and offshore structures.
CAD	Computer-Aided Design , Software used to create, modify, analyse, and optimise 2D or 3D engineering designs (e.g., CATIA, NX, SolidWorks).
CATALYST	Change Analysis Through Linked Systems and Intelligent Tracking , The proposed Engine System name and acronym for this submission.
Celery	An open-source distributed task queue for Python used to dispatch and manage asynchronous background jobs (e.g., ML inference tasks).
CI	Confidence Interval , A statistical range around a predicted value within which the true value is expected to fall with a specified probability (e.g., 80% or 95% CI).
CPM	Critical Path Method , A project scheduling technique that identifies the longest sequence of dependent tasks determining the minimum project duration.



Abbreviation / Term	Definition / Explanation
CSV	Comma-Separated Values , A plain-text file format for tabular data where each field is separated by a comma; used for BOM ingestion.
DAG	Directed Acyclic Graph , A graph data structure with directed edges and no cycles; used to represent project schedules in the PERT/CPM module.
Digital Twin	A virtual, data-driven replica of a physical asset (here, a ship component) that mirrors its geometric, material, mass, and BOM properties in real time.
DNV	Det Norske Veritas , A Norwegian classification society issuing design and safety standards for maritime and offshore industries.
Docker / Docker-Compose	A containerisation platform (Docker) and multi-container orchestration tool (Docker-Compose) that package applications and their dependencies into portable, isolated units for on-premise deployment.
Drools	An open-source Business Rules Management System (BRMS) by Red Hat used to encode, manage, and execute deterministic engineering constraint validation rules.
ERP	Enterprise Resource Planning , Integrated business management software (e.g., SAP PM, Oracle) that centralises procurement, finance, and supply-chain data including BOM cost rates.
FastAPI	A modern, high-performance Python web framework for building REST APIs; used as CATALYST's async backend API gateway.
FOM	Field Operations Manual , A reference document defining standard operating procedures for system maintenance and operational processes (here, the rule update process).
IEC	International Electrotechnical Commission , Global standards body for electrical and electronic technologies; IEC 61084-1 (cable tray fill ratios) and IEC 60092 (cable bend radius) are referenced in CATALYST constraints.
IRS	Indian Register of Shipping , India's national ship classification society responsible for technical standards and certification of Indian-flagged vessels.
JSON	JavaScript Object Notation , A lightweight, human-readable data interchange format used for structured API payloads and change schema definitions in CATALYST.
LDAP	Lightweight Directory Access Protocol , A protocol for accessing and maintaining distributed directory information services (e.g., user accounts) over an IP network; used for on-premise RBAC integration.



Abbreviation / Term	Definition / Explanation
MAWP	Maximum Allowable Working Pressure , The maximum safe operating pressure a pressurised system component (e.g., piping) is designed to withstand per its specification.
MinIO	An open-source, high-performance object storage server compatible with the Amazon S3 API; used in CATALYST for on-premise storage of ML model artifacts and large binary files.
ML	Machine Learning , A subset of AI in which algorithms learn patterns from data to make predictions or decisions without explicit rule programming.
mTLS	Mutual Transport Layer Security , A security protocol extension in which both client and server authenticate each other's certificates; used in CATALYST's Phase 6 zero-trust microservices layer.
Neo4j	A leading native graph database that stores data as nodes and relationships; used as CATALYST's Dependency Knowledge Graph engine, queried via the Cypher language.
NER	Named Entity Recognition , An NLP sub-task that identifies and classifies key entities in text (e.g., component names, change types, magnitudes) into predefined categories.
NetworkX	A Python library for creating, analysing, and manipulating complex network graphs; used in the PERT/CPM module to compute critical path and float consumption.
NLP	Natural Language Processing , A field of AI focused on enabling computers to understand, interpret, and generate human language text or speech.
OCC / OpenCASCADE	Open CASCADE Technology , An open-source geometric modelling kernel for 3D CAD, CAM, and CAE; Optional CAD geometry kernel used in Phase 6 enhancements for 3D assembly ingestion; not required for core PARINAM operation.
OData	Open Data Protocol , A standardised REST-based protocol for querying and updating data; used in CATALYST's ERP connector (SAP/Oracle) for BOM and cost data synchronisation.
PERT	Program Evaluation and Review Technique , A statistical scheduling method that uses weighted-average (optimistic, most likely, pessimistic) durations to estimate task completion time under uncertainty.
PLM	Product Lifecycle Management , Software platforms (e.g., Siemens Teamcenter, PTC Windchill) that manage the entire lifecycle of a product from design through decommissioning, including revision control and BOM management.



Abbreviation / Term	Definition / Explanation
PostgreSQL	An open-source, enterprise-grade relational database management system; used in CATALYST for persistent storage of Digital Twin node data, audit logs, and change history.
PyTorch	An open-source deep learning framework developed by Meta; used for training and fine-tuning neural network models (e.g., BERT fine-tuning in CATALYST).
RBAC	Role-Based Access Control , A security model that restricts system access based on a user's assigned role (Viewer, Analyst, Approver, Admin in CATALYST).
React / TypeScript	React is a JavaScript library for building interactive user interfaces; TypeScript is a statically typed superset of JavaScript. Together they form the stack for CATALYST's Decision Intelligence Dashboard.
Redis	An open-source in-memory data store used as a message broker and task queue backend for Celery in CATALYST's asynchronous job dispatch layer.
REST	Representational State Transfer , An architectural style for designing networked APIs that use standard HTTP methods (GET, POST, PUT, DELETE) for resource operations.
SAP PM / Oracle	SAP Plant Maintenance (SAP PM) and Oracle are enterprise ERP platforms widely used in defence and industrial organisations for maintenance management and procurement; CATALYST integrates with these for BOM and cost data.
SHAP	SHapley Additive exPlanations , A game-theoretic ML interpretability framework that attributes each feature's contribution to a model's prediction; used in CATALYST to explain top risk drivers in plain English.
SME	Subject Matter Expert , A domain specialist (e.g., a naval architect or systems engineer) whose knowledge is captured in the dependency matrix and PERT duration estimates.
UUID	Universally Unique Identifier , A 128-bit label used to uniquely identify objects (here, each Digital Twin component node) across distributed systems without collision risk.
Uvicorn	An ASGI-compliant web server implementation for Python used to run FastAPI applications with high concurrency and async I/O performance.
XML	Extensible Markup Language , A hierarchical, human-readable data format for structured document exchange; used for BOM ingestion and Primavera P6 schedule export in CATALYST.



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G1603, SAMARTH GARDEN, BHANDUP (W), MUMBAI 400078, MAHARASHTRA.
9137815600 / 9029767757
pm.team@arcaisys.com

Abbreviation / Term	Definition / Explanation
XER	Primavera P6 Export format , A proprietary file format used by Oracle Primavera P6 to export and import project schedule data including task sequences, durations, and resource assignments.
XGBoost	Extreme Gradient Boosting , A high-performance, regularised gradient boosting ML algorithm; used in CATALYST's Layer 2 cost estimation model to predict total change cost with confidence intervals.
Zero-Trust Architecture	A cybersecurity model that assumes no implicit trust for any user, device, or network segment , every access request is continuously verified. Implemented in CATALYST Phase 6 via mTLS between microservices and RBAC combined controls.